University of Ghana
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Project Report

SleepWise

A Sleep Quality Advisor Expert System for University Students

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Abstract

SleepWise is a Knowledge-Based System we built to help university students work out why their sleep is poor. The system asks between 20 and 50 questions depending on which session variant the student chooses, then gives them a personalised report with the causes of their sleep problems and specific steps to fix each one.

The diagnostic engine has two parts. SWI-Prolog holds all the intelligence ((29 cause rules, a four-level severity calculator, six sleep complaint types, five clinical alert flags, and 29 recommendations). Python handles the interface, a Bayesian probability engine that chooses which questions to ask next, and the session management. The two halves talk to each other through pyswip.

We tested the system with 267 automated test cases, all of which passed. We also ran a large-scale simulation of 951 student sessions to check for edge cases. This report covers the full project — the problem, the system design, the knowledge base, the implementation, and the results.

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1. Introduction

1.1 The Problem

Sleep is a real problem for university students in Ghana. A 2022 study by Mbous et al. looked at students across sub-Saharan Africa and found that 35% showed signs of depression that were directly affecting their sleep. 55% were dealing with a combined academic and workload heavy enough to cause chronic sleep disruption. For students at the University of Ghana, living conditions, financial stress, and exam pressure add extra layers on top of those general patterns.

The hard part is that sleep problems rarely come from just one thing. A student might be drinking too much coffee, studying past midnight, living in a noisy room, and feeling anxious about exams all at the same time. A system that only checks one or two factors at a time will miss the full picture.

Most students dealing with sleep problems do not see a doctor or counsellor. The cost and time involved makes that impractical. But if a student could answer a short set of questions and get a personalized explanation of what is affecting their sleep with specific steps to fix each problem that could actually help.

1.2 What We Built

SleepWise is an expert system that diagnoses sleep problems for university students. It has four session variants the student can choose from: a 20-question quick session, a 30-question standard session, a 40-question detailed session, and a 50-question full session. The longer variants ask more follow-up questions and can detect a wider range of causes.

After the session ends the student gets a full report. It lists which causes were found, how serious the overall problem is, what type of sleep complaint they have, and a specific recommendation for each cause. If the combination of answers suggests something that needs professional help — like severe depression with short sleep, or nightly panic attacks — the system flags that and directs the student to campus counselling.

1.3 Why a Knowledge-Based System

We chose a Knowledge-Based System approach because sleep diagnosis is rule-based. Doctors and researchers have spent decades building clinical tools the Pittsburgh Sleep Quality Index, the Insomnia Severity Index, the DASS-21 that turn sleep-related questions into diagnostic conclusions using fixed rules and thresholds. That is exactly the kind of reasoning a KBS is good at.

We used SWI-Prolog to store all the rules. Prolog is designed for symbolic reasoning, you write a rule, and when the facts match, the conclusion fires. This kept the intelligence inside the knowledge base, separate from the Python code that handles the interface. A marker can open sleepwise.pl and read the diagnostic logic directly. It is transparent and verifiable.

1.4 Report Structure

Section 2 covers background — how expert systems work and what research already exists on student sleep. Section 3 explains the architecture. Section 4 covers the knowledge base in detail.

Section 5 covers the implementation. Section 6 presents the test results. Section 7 discusses what worked and what the limitations are. Section 8 is the conclusion.

2. Background and Related Work

2.1 Expert Systems

An expert system is a computer program that uses stored rules to make decisions the way a human expert would.

Prolog is one of the standard languages for building expert systems. For our system, we ask sleep_cause(C) and Prolog checks all 29 cause rules to see which ones fire.

2.2 Sleep Problems in Students

Several large studies have looked specifically at university students. Okano et al. (2019) tracked 100 college students with wearable sensors and found that sleep consistency — going to bed and waking at the same time mattered more for academic performance than total sleep hours. 60% of students in that study had inconsistent schedules.

Wang et al. (2023) found that 70% of students report high stress levels and 45% show poor routine adherence. They also showed that stress and short sleep form a two-way cycle each makes the other worse.

Bousgheiri et al. (2024) studied North African students — the closest geographic group to the University of Ghana with available data — and found that 72% use screens right before bed. This was the highest prevalence finding across all our sources, which is why excessive_screen_use has the highest prior probability in our system.

2.3 Clinical Measurement Tools

We based our diagnostic logic on three validated clinical instruments.

The Pittsburgh Sleep Quality Index (PSQI) measures seven components of sleep quality: subjective quality, latency, duration, efficiency, disturbances, medication use, and daytime function. It has a threshold of 30 minutes for sleep latency (anything longer is a problem) and a 7-hour boundary for sleep duration.

The Insomnia Severity Index (ISI) separates insomnia into three types: difficulty falling asleep, waking during the night, and daytime impairment. Our three complaint rules for insomnia, maintenance_insomnia, and performance_impact map directly to these three ISI categories.

The Depression Anxiety Stress Scales (DASS-21) treats depression, anxiety, and stress as three separate but related constructs. This is why our system has three independent cause rules for them rather than one general psychological distress rule, the DASS-21 evidence shows they have different symptom patterns and different interventions.

2.4 Existing Sleep Apps

There are apps like Sleep Cycle, Calm, and Headspace that track sleep or help people relax. But none of them diagnose why sleep is poor. They measure or improve sleep without identifying the cause. SleepWise is different it is a diagnostic tool, not a tracking tool. The output is a causal explanation with specific advice, not just a sleep score.

3. System Design and Architecture

3.1 Overall Structure

SleepWise has two main components: a Prolog knowledge base and a Python application layer. They are connected through pyswip, a Python library that lets Python code call SWI-Prolog predicates.

Here is how a session works, step by step:

1. The student runs interface.py and picks a session variant (q20, q30, q40, or q50).
2. The session_manager.py selects the first batch of questions from question_bank.py.
3. For each answer, Python runs assertz(user_fact(question_id, answer_value)) to put that fact into Prolog.
4. The bayes_engine.py updates posterior probabilities for all 29 causes after each answer.
5. trigger_rules.py decides whether to add follow-up questions based on the current posteriors.
6. When all questions are done, interface.py queries Prolog for sleep_cause(C), sleep_severity(S), sleep_complaint(C), sleep_clinical_flag(F, L), and recommendation(Cause, Rec).
7. The results are displayed to the student as a formatted report.
8. Python calls reset_session/0 to clear all user_fact assertions before the next student.

3.2 File Structure

File	What it does
sleepwise.pl	The Prolog knowledge base. All 29 cause rules, severity, complaints, clinical flags, and recommendations.
bayes_engine.py	Naive Bayes inference engine. Computes posterior probabilities for all 29 causes after each answer.
question_bank.py	All 120 questions with their IDs, answer options, and variant assignments.
session_manager.py	Controls question flow, calls the Bayes engine, manages the session state.
trigger_rules.py	Rules for when to add extra questions based on high-probability causes.
interface.py	Terminal interface. Asks questions, shows the report, queries Prolog.

3.3 The Python-Prolog Bridge

The bridge between the two halves of the system is a single dynamic predicate in Prolog:

```
:- dynamic user_fact/2.
```

Python asserts one user_fact clause for each question the student answers. For example, if a student says their stress level is high:

```
prolog.assertz('user_fact(stress_level, high)')
```

Then when we query sleep_cause(C), the Prolog rule for high_stress checks whether user_fact(stress_level, high) exists. It does, so the rule fires. At the end of the session:

```
prolog.retractall('user_fact(_, _)')
```

This clears everything so the next student starts clean.

3.4 Four Session Variants

Variant	Core questions	Max with add-ons	Who it is for
q20	20	29	Student who wants a quick check
q30	30	39	Standard session for most students
q40	40	49	Detailed assessment
q50	50	59	Full assessment — includes sleep disorder questions

Add-on questions are triggered by the Bayesian engine when a cause reaches high enough posterior probability mid-session. Clinical questions (about panic attacks, self-harm risk, diagnosed disorders) are only included in the longer variants.

3.5 The Bayesian Engine

The Bayesian engine starts with a prior probability for each of the 29 causes. These priors come from prevalence data in the research literature — for example, excessive_screen_use starts at 0.72 because Bousgheiri 2024 found 72% of African students use screens before bed.

After each answer, the engine updates all 29 posteriors using Naive Bayes in log-space. Log-space prevents underflow when multiplying many small probabilities together. After updating, it applies a correlation correction to eight pairs of questions that are known to co-occur frequently (like stress and rumination, or screen use and irregular schedule). This partially compensates for the independence assumption that Naive Bayes makes.

The final posteriors are normalised so they sum to 1, then compared against a distribution-relative threshold. Causes that are more than one standard deviation above the mean posterior are flagged

as fired. If fewer than three causes fire and the anchor questions suggest a real sleep problem, the threshold drops slightly to catch borderline cases.

4. Knowledge Base

4.1 How the Rules Are Written

Every cause rule in sleepwise.pl follows the same pattern:

```
sleep_cause(cause_name) :-
    user_fact(question_id, Value),
    member(Value, [problem_values]).
```

The question_id must exactly match a question ID in question_bank.py. The answer values are atoms — qualitative words like 'often', 'high', 'diagnosed_severe'. We used member/2 for set membership rather than arithmetic because our answers are categories, not numbers. This is the correct approach for a symbolic knowledge base.

Prolog's open-world assumption handles missing questions naturally. If a question was not asked in a given session, user_fact simply fails and the rule does not fire. No special handling is needed.

4.2 The 29 Causes

The 29 causes are split across five domains:

Domain	Causes	Cause names
Lifestyle	6	high_caffeine, poor_diet_habits, sedentary_lifestyle, substance_use, poor_hydration, medication_interference
Psychological	8	high_stress, anxiety, depression, burnout, rumination, poor_self_control, perfectionism, low_resilience
Environmental	5	noise_disturbance, light_disturbance, temperature_discomfort, poor_sleep_space, safety_concern
Behavioural	5	excessive_screen_use, irregular_schedule, poor_sleep_hygiene, excessive_napping, poor_sleep_efficiency
Academic	5	poor_sleep_duration, academic_overload, late_night_studying, exam_pressure, performance_impact

Some causes have two separate Prolog clauses because they have two independent pathways. Substance use is the clearest example — alcohol problems and drug problems are both substance use issues, so either clause alone fires the same cause:

```
sleep_cause(substance_use) :-
    user_fact(alcohol_consumption, Value),
    member(Value, [often, daily]).
```

```
sleep_cause(substance_use) :-
    user_fact(recreational_drug_use, Value),
    member(Value, [often, frequently]).
```

4.3 Severity Levels

The severity predicate has four levels. Severe fires in three situations: four or more causes are found, depression is present alongside under four hours of sleep per night, or the student has a diagnosed severe sleep disorder:

```
sleep_severity(severe) :-
    (   findall(C, sleep_cause(C), Cs), length(Cs, N), N >= 4
    ;   user_fact(sleep_hours, V), member(V, [less_than_4, four_to_six]),
        user_fact(depression_symptoms, D), member(D, [often, every_day])
    ;   user_fact(sleep_disorders, diagnosed_severe)
    ), !.
```

The four-cause threshold comes from the PSQI global score structure — a PSQI score above 10 (severe) typically reflects problems in at least four of the seven PSQI components. The depression plus short sleep combination comes from Mbous 2022, which identified that pattern as the highest-risk combination in sub-Saharan African students.

Severity level	Condition
severe	4+ causes fired, OR depression + under 4 hours sleep, OR diagnosed severe disorder
moderate	2-3 causes fired, OR short sleep (4-6 hours) without depression
mild	1 cause fired
none	No causes fired

4.4 Sleep Complaints

Complaints are the plain-English description of the student's sleep problem. A student can have more than one. There are six:

Complaint	What triggers it
sleep_deprivation	sleep_hours in [less_than_4, four_to_six]
insomnia	sleep_latency in [slow, very_slow]
maintenance_insomnia	frequent_wakeups in [several, many]
hypersomnia	daytime_nap_hours in [long] AND morning_sleepiness in [groggy, extremely_tired]

Complaint	What triggers it
circadian_disruption	sleep_time = inconsistent AND screen_before_bed in [often, always]
stress_related_insomnia	stress_level = high AND sleep_latency in [slow, very_slow]

4.5 Clinical Alert Flags

The five clinical flags fire when the answers suggest something beyond what sleep hygiene advice can fix. Every flag requires multiple conditions to be true at the same time — no single answer alone can trigger a referral:

Flag	Conditions required	Source
CF1	Depression often/every_day AND sleep under 4 hours	Mbous 2022
CF2	Anxiety every night AND panic attacks often/frequently	ISI / DASS-21
CF3	Diagnosed severe sleep disorder	ISI threshold
CF4	Burnout constantly AND depression often/every_day	Stores 2023 / Mbous 2022
CF5	Depression + anxiety + sleep under 6 hours — all three together	Wang 2023 / DASS-21

4.6 Recommendations

Every cause has exactly one recommendation stored as a Prolog fact. The recommendations are specific. For example, the noise_disturbance recommendation says to use NRR-33 foam earplugs, because AASM guidelines identify that specific product category as the most evidence-backed noise intervention — it reduces ambient sound by about 20dB, which is enough to block most conversational and traffic noise.

All 29 recommendations live in sleepwise.pl, not in Python. This is an important design choice — the intelligence belongs in the knowledge base.

4.7 Bugs Fixed During Development

During testing we found and fixed seven bugs. All seven were in the Prolog knowledge base or the Bayesian engine — none were interface bugs.

Bug	Location	Problem	Fix
1	sleepwise.pl	substance_use used 'regularly' — not a valid atom in the question bank	Changed to 'frequently'

Bug	Location	Problem	Fix
2	sleepwise.pl	poor_self_control used [often, always] — wrong answer type	Changed to [moderate, severe]
3	sleepwise.pl	excessive_screen_use used [one_to_two_hours, over_two_hours] — phantom atoms	Changed to [moderate, long]
4	sleepwise.pl	irregular_schedule used [much_longer, very_much_longer] — phantom atoms	Changed to [much_more, far_more]
5	sleepwise.pl	performance_impact fired on [often, always] — backwards logic	Changed to [never, rarely] for attendance
6	sleepwise.pl	hypersomnia used [often, always] — wrong atom type for morning sleepiness	Changed to [groggy, extremely_tired]
7	bayes_engine.py	high_stress likelihood for stress=medium was 0.55 — caused over-diagnosis	Changed to 0.25

5. Implementation

5.1 Technology Stack

Component	Technology
Knowledge base	SWI-Prolog — sleepwise.pl (606 lines)
Bayesian engine	Python 3 — bayes_engine.py (1,865 lines)
Question bank	Python 3 — question_bank.py (2,185 lines, 120 questions)
Session manager	Python 3 — session_manager.py (483 lines)
Trigger rules	Python 3 — trigger_rules.py (788 lines)
Interface	Python 3 — interface.py (1,085 lines), terminal-based
Bridge	pyswip — Python library for calling SWI-Prolog
Total codebase	7,012 lines across 6 files

5.2 The Question Bank

The question bank has 120 questions covering all five domains. Each question has a unique ID, a text for the student, a list of answer options as atoms, and a list of which session variants include it. Some questions only appear in the longer variants — for example, questions about diagnosed sleep disorders only appear in q50.

The answer atoms match exactly to what the Prolog rules expect. This matching is how the bridge between Python and Prolog works — Python asserts user_fact(question_id, answer_atom) and Prolog rules check for specific answer atoms by name.

5.3 Adaptive Question Selection

The system does not ask the same questions to every student in a fixed order. After each answer, the Bayesian engine updates all 29 cause posteriors. The trigger_rules.py file then checks whether any cause has reached a threshold high enough to justify asking follow-up questions about it.

For example, if anxiety reaches a high posterior after the student answers that they feel anxious often, the system adds the panic attacks question as an add-on. This is how the system stays within its question budget while still gathering targeted information for high-probability causes.

5.4 The Diagnosis Report

The final report has four sections. First, any clinical flag is shown prominently at the top in red with a message directing the student to counselling. Second, the overall severity level is shown. Third, each fired cause is listed with the cause name, its domain, the confidence level, a plain-English description of what the cause is, how it affects sleep, and a specific recommendation. Fourth, the sleep complaint type is shown at the bottom.

Each cause listing also shows a [Prolog check] tag if Prolog confirmed the same cause that the Bayesian engine found. This cross-checking happens in real time — the system asks Python and Prolog independently and notes where they agree.

5.5 Graceful Degradation

If Prolog is unavailable for any reason, the system falls back to Python-only mode. It still runs the Bayesian engine, still produces a severity score, still shows complaints, and still gives recommendations from a Python fallback dictionary. The student gets a complete report — it just does not have the Prolog cross-check tags.

6. Testing and Results

6.1 Test Strategy

We ran three types of testing. First, unit tests on the Prolog knowledge base in isolation. Second, integration tests on the full Python-Prolog pipeline. Third, a large-scale simulation of 951 complete student sessions.

6.2 Knowledge Base Tests — 189/189 passed

The test script test_knowledge_base.py ran 189 tests directly against sleepwise.pl using pyswip. Every test passed.

Test section	Tests	Passed	Result
All 29 causes fire on correct inputs	81	81	PASS
Healthy/below-threshold values produce no cause	38	38	PASS
All four severity levels	6	6	PASS
All 5 clinical flags — positive and negative	20	20	PASS
All 6 sleep complaint types	7	7	PASS
All 29 recommendations exist and are non-empty	29	29	PASS
Session reset and isolation	5	5	PASS
TOTAL	**189**	**189**	**PASS**

The negative test cases were as important as the positive ones. For example, we tested that stress_level=medium does not fire high_stress (Bug 7 fix), that lecture_attendance=often does not fire performance_impact (Bug 5 fix), and that a healthy student with all good answers gets zero causes and severity=none.

6.3 Integration Tests — 78/78 passed

The test script test_system_integration.py ran 78 tests on the full pipeline. Every test passed.

Test section	Tests	Passed	Result
Prolog bridge initialises correctly	5	5	PASS
Python assert → Prolog query round-trip	5	5	PASS
Python and Prolog agree on 10 causes	12	12	PASS
Severity levels match across both engines	9	9	PASS
All 29 recommendations retrievable through bridge	12	12	PASS
Full session pipeline — all 4 variants	20	20	PASS

Test section	Tests	Passed	Result
Multi-session isolation — no fact bleed	6	6	PASS
Graceful degradation — Python-only mode	9	9	PASS
TOTAL	**78**	**78**	**PASS**

The agreement tests between Python and Prolog required careful input design. Prolog fires a cause when a single matching fact is asserted. The Python Bayesian engine needs multiple corroborating signals to push a cause above the threshold. So the agreement test inputs provided all the relevant likelihood questions at triggering values — not just one.

6.4 Large-Scale Simulation — 951 sessions

We ran simulate_all.py to test the system across 951 complete student sessions covering a wide range of student profiles. The simulation created different student personas — high-stress exam students, students in noisy accommodation, students with lifestyle issues — and ran them through all four session variants.

The simulation confirmed the system handles edge cases correctly. The most important finding was that a genuinely healthy student with all good answers gets zero fired causes and severity=none. This sounds obvious but it is actually a hard test — the Bayesian engine's prior probabilities mean some causes start at 0.70 before any answers come in, so if the fallback threshold is too loose, healthy students get incorrectly diagnosed.

6.5 Summary

Test type	Tests run	Result
Knowledge base unit tests	189	189/189 PASS
System integration tests	78	78/78 PASS
Large-scale simulation	951 sessions	Completed — edge cases verified
Total automated tests	267	267/267 PASS

7. Discussion

7.1 What Worked Well

The separation between Prolog and Python turned out to be the right call. Prolog rules are readable by the marker directly. You can open sleepwise.pl and see exactly what conditions trigger each cause. That transparency makes the system verifiable in a way that a hidden machine learning model would not be.

The adaptive question selection also worked better than expected. Instead of asking every student all 50 questions, the system focuses on high-probability areas. A student with no stress signals skips all the psychological follow-up questions. This keeps the session short and focused.

Bug 7 — the high_stress prior miscalibration — was the most impactful fix. Before the fix, any student who said their stress level was 'medium' got high_stress flagged as a cause. After the fix, only 'high' stress triggers the cause. That change alone fixed overdiagnosis across a large fraction of sessions.

7.2 Limitations

The system relies on self-reported answers. Students may underreport sensitive issues like depression or drug use. We partially addressed this with self-report bias corrections in the likelihood tables — the likelihoods for sensitive questions assume some underreporting — but this cannot fully solve the problem.

The Bayesian prior probabilities were set using prevalence data from the literature, not from a dataset of University of Ghana students specifically. The Bousgheiri 2024 and Mbous 2022 data are the closest available geographic proxies, but they are not exact.

The system is a diagnostic tool, not a treatment system. It can tell a student that anxiety is affecting their sleep and recommend breathing exercises. But if the anxiety is clinical, only a professional can provide actual treatment.

7.3 KBS Versus Machine Learning

We could have built SleepWise using a machine learning classifier instead of a knowledge base. A decision tree or neural network trained on student sleep data might reach the same conclusions. But there are two reasons we prefer the KBS approach for this problem.

First, explainability. When a KBS diagnoses high_stress as a cause, you can point to the rule that fired and the fact that matched it. A student can see exactly why the system reached that conclusion. A black-box classifier cannot do that.

Second, data. Training a reliable classifier for student sleep problems in Ghana would require a labelled dataset of thousands of student sessions with confirmed diagnoses. That data does not exist. The KBS approach lets us use the clinical rules that sleep researchers have already validated — the PSQI, ISI, and DASS-21 — directly.

7.4 Future Work

The most useful improvement would be a web or mobile interface. The current terminal interface is fine for a DCIT 313 project but would not be practical for real students to use.

Adding a longitudinal feature — letting a student repeat the assessment after a few weeks to see whether their sleep improved — would make the recommendations more useful.

Collecting anonymised session data with student consent would allow calibration of the prior probabilities specifically for University of Ghana students, making the Bayesian engine more accurate for this population.

8. Conclusion

SleepWise shows that a Knowledge-Based System built on validated clinical rules can produce accurate, explainable sleep diagnoses for university students. The Prolog knowledge base holds 29 cause rules derived from the PSQI, ISI, DASS-21, and four peer-reviewed studies on student sleep. The Python layer handles question selection, Bayesian probability estimation, and the session interface.

All 267 automated tests passed. The system correctly diagnoses causes when they are present and stays silent when conditions are healthy. The seven bugs found during development were all in the knowledge base or likelihood tables — not in the interface — which shows that the testing process worked as intended.

The main contribution of this project is a transparent, testable, source-traceable diagnostic system. Every rule can be traced back to a clinical source. Every test case documents expected and actual behaviour. The full test scripts are included in the repository so any marker can run them and verify the results live.

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